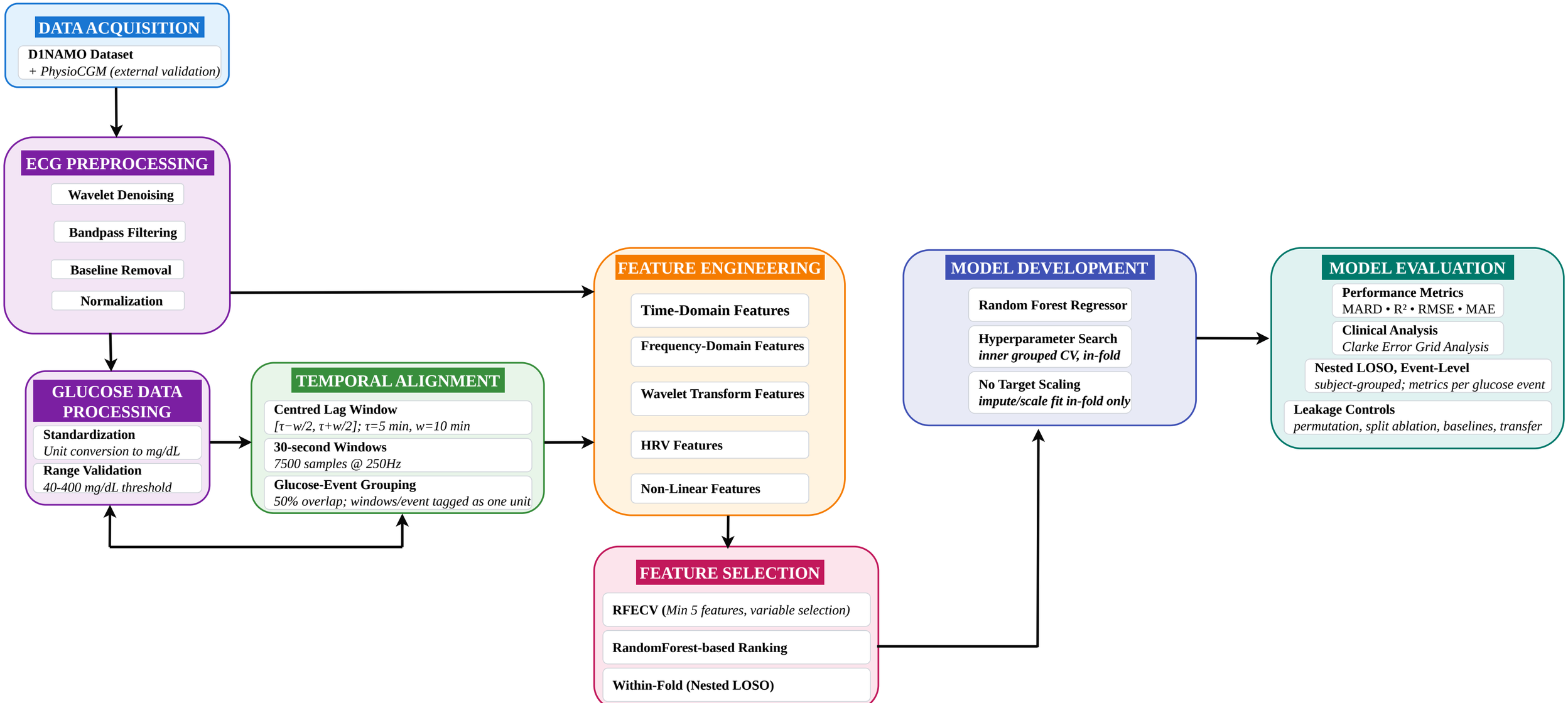

DATA ACQUISITION
D1NAMO Dataset
+ PhysioCGM (external validation)
ECG PREPROCESSING
Wavelet Denoising
Bandpass Filtering
Baseline Removal
Normalization
GLUCOSE DATA PROCESSING
Standardization
Unit conversion to mg/dL
Range Validation
40-400 mg/dL threshold
TEMPORAL ALIGNMENT
Centred Lag Window
[τ−w/2, τ+w/2]; τ=5 min, w=10 min
30-second Windows
7500 samples @ 250Hz
Glucose-Event Grouping
50% overlap; windows/event tagged as one unit
FEATURE ENGINEERING
Time-Domain Features
Frequency-Domain Features
Wavelet Transform Features
HRV Features
Non-Linear Features
FEATURE SELECTION
RFECV (Min 5 features, variable selection)
RandomForest-based Ranking
Within-Fold (Nested LOSO)
MODEL DEVELOPMENT
Random Forest Regressor
Hyperparameter Search
inner grouped CV, in-fold
No Target Scaling
impute/scale fit in-fold only
MODEL EVALUATION
Performance Metrics
MARD • R² • RMSE • MAE
Clinical Analysis
Clarke Error Grid Analysis
Nested LOSO, Event-Level
subject-grouped; metrics per glucose event
Leakage Controls
permutation, split ablation, baselines, transfer